



Top-down vs. bottom-up synthesis of Ga-based supported catalytically active liquid metal solutions (SCALMS) for the dehydrogenation of isobutane

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ABSTRACT

Supported catalytically active liquid metal solutions (SCALMS) catalysts consist of a liquid metal solution of an active metal (e.g., Pd, Rh, Pt, Ni) in a low melting metal (e.g., Ga, In, Sn) deposited on a porous support material. In the current study, we extend and optimize the recently developed top-down synthesis protocol (ultrasonication of gallium metal) to prepare GaPd, GaRh and GaPt SCALMS systems. We compare the morphological differences in the obtained catalyst materials achieved between this approach and the more traditional bottom-up synthesis protocol (*in situ* thermal decomposition of triethylamino-gallane complex) using electron microscopy. Moreover, the spatial distribution of the catalytically active material in the porous alumina support is evaluated using high resolution argon physisorption measurements. Both techniques strongly indicate that the bottom-up approach leads to a high degree of pore penetration while the gallium droplets are largely on the outer surface in case of the top-down method. Changes to the surface chemistry of the support were also investigated by means of NH₃/CO₂ temperature programmed desorption. All catalytic systems synthesized by the two synthesis methods, i.e., GaPd, GaRh and GaPt, were tested in a quartz, fixed-bed tubular reactor for isobutane dehydrogenation. All SCALMS materials, irrespective of their individual synthesis protocol, were significantly more active than their pure metal counterparts. The SCALMS catalyst prepared using the bottom-up approach showed initial productivity of 407 g_{butenes} mol_M⁻¹ min⁻¹, 832 g_{butenes} mol_M⁻¹ min⁻¹, and 963 g_{butenes} mol_M⁻¹ min⁻¹ for the GaPd, GaRh and GaPt respectively. These values represent more than twice the initial activity obtained with the top-down synthesized counterparts. This activity difference can be explained by the analytically determined difference in liquid metal droplet size distribution.

1. Introduction

Short chain alkenes, especially propene and butene, are among the most important platform chemicals due to their versatility as chemical building blocks for various products as well as intermediates [1]. Currently, these compounds are mostly produced via steam cracking and fluid catalytic cracking (FCC) of larger hydrocarbons [2]. However, a steady increase in demand has raised interest in on-purpose production of these alkenes by the dehydrogenation of their respective alkanes [3]. One of the major industrial challenges of these alkane dehydrogenation

processes, is the rapid deactivation of the catalyst by coking. All industrial processes currently in industrial use either have short reaction cycles followed by regeneration of the catalyst or employ an oxygen/steam assisted dehydrogenation process that drastically affects the selectivity of the process [1].

To suppress catalyst deactivation by coke formation, the addition of ‘catalyst poisons’ has been explored to inhibit the most active catalyst sites which are most prone to coke formation [2,4]. Also, bimetallic catalysts, such as PtSn or PtGa, and single atom catalysts (SAC), are known to drastically improve the catalyst performance and inherently

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inhibit coke formation [5]. We recently introduced a novel approach to achieve single atom catalysts in a technical manner by using supported catalytically active liquid metal solutions (SCALMS). Herein, the active metal is mixed with a low melting metal such as gallium and the formed liquid metal solution is dispersed on a porous support [6]. In SCALMS, the composition of the metal mixture is adjusted based on the inter-metallic phase diagram to ensure complete dissolution of the active metal in gallium under reaction conditions [7–10,11]. Ga-based SCALMS catalysts with Pd, Rh and Pt have been tested in butane, propane and methylcyclohexane dehydrogenation and more recently, non-noble metal GaNi SCALMS have been also reported as an active catalyst for ethylene oligomerization [6,11–14]. In all these applications, SCALMS showed significant resistance to coke formation and higher activity compared to the corresponding monometallic, solid metal counterparts (here referred to as bulk metal catalysts). Interestingly, GaRh SCALMS showed exceptional activity even though supported Rh alone is an extremely poor catalyst for propane dehydrogenation, indicating advantageous electronic effects caused by the liquid alloy character of SCALMS [14–16].

SCALMS synthesis proceeds via deposition of gallium onto the support surface, followed by galvanic displacement to introduce the active metal into the Ga matrix. We have explored two drastically different synthesis routes for the deposition of gallium on the selected support. GaPd and GaRh SCALMS catalyst were synthesized by the thermal decomposition of triethylamino-gallane, $(\text{Et}_3\text{N})\text{GaH}_3$ (bottom-up approach) [6,14,17]. This synthesis protocol involves highly sensitive and hazardous chemicals, is not very atom efficient and hence less suitable for scale-up. An alternative route was therefore developed, where gallium droplets are formed by the ultrasonication of gallium nuggets in a solvent followed by deposition on a support (top-down approach) [18–20]. The process involves fewer steps, is 100% atom efficient with complete recycling of the applied solvent and is easier to scale up [13]. More recently, direct dissolution of platinum in bulk gallium was reported to synthesize Ga-Pt liquid metal solution as highly active catalysts [21]. This approach clearly demonstrated the high activity of a fully liquid GaPt metal solution for multiple low temperature reactions. However, the latter synthesis method is not amenable for supported systems with much higher loading of Pt that still can form supported liquid alloys at high reaction temperatures.

In the current study, GaPd, GaRh and GaPt SCALMS systems were synthesized by the bottom-up as well as the top-down route. We here compare in detail the difference in morphology and spatial distribution of the metal droplets using electron microscopy and high-resolution Ar-physisorption. We then correlate the catalytic activity of the two differently prepared catalysts in isobutane dehydrogenation in comparative studies under identical reaction conditions. This will allow to assess the efficiency of the more benign ultrasonication method and hence can be a baseline for future scale-up studies. Note, that isobutane dehydrogenation is thermodynamically more favorable than ethane, propane, or *n*-butane dehydrogenation because of the weak tertiary C-H bond. Thus, the equilibrium conversion for isobutane dehydrogenation at a given temperature and pressure is higher than that for the dehydrogenation of these other C_2 – C_4 alkanes. The product of isobutane dehydrogenation, isobutylene, is of industrial relevance to produce ethyl *tert*-butyl ether, fuel additives and butyl rubber amongst many other applications [22–24]. This makes isobutane dehydrogenation a very suitable reaction to compare the activity, selectivity and deactivation behavior of catalyst materials for alkane dehydrogenation [25].

2. Materials and methods

All chemicals were used as received. Standard Schlenk techniques were used for the preparation of SCALMS with Ar (5 N purity, Linde Gas) as inert gas. Alumina (Sigma-Aldrich; grade: Brockmann I, activated, standard; particle size: 0.05–0.15 mm; BET surface area: $155 \text{ m}^2 \text{ g}^{-1}$; pH: 7.0 ± 0.5) was used as the support material for all catalysts.

2.1. Preparation of catalysts

For the top-down approach using ultrasonication (indicated by ‘td’ in the catalyst ID), supported Ga catalysts were prepared by physical deposition of metallic gallium on the porous support. A 6 mm Gallium nugget, weighing approximately 0.8–0.9 g (5 N, Alpha-Aesar) was ultrasonicated in 100 mL propan-2-ol using a Branson 450 sonicator (80% intensity for 10 min) at 313 K creating an emulsion of liquid Ga droplets. The required amount of support material to achieve a 5% loading was subsequently added to the emulsion and the solvent was slowly evaporated in a rotary evaporator at 323 K. The obtained Ga-decorated support was calcined at 723 K for 2 h (Fig. 1 (td)). In case of the bottom-up approach for Ga deposition (indicated by ‘bu’), Ga was added to the support using the procedure described by Bauer et al. [26]. The procedure involves the *in situ* formation of triethylamino-gallane by the reaction of gallium chloride with triethylamino hydrochloride and lithium hydride. The triethylamino-gallane readily decomposes to metallic gallium at elevated temperatures (Fig. 1 (bu)).

In all cases, the final SCALMS catalyst was prepared by the addition of the active metal i.e., Pd, Rh and Pt, via galvanic displacement (Fig. 1 (GD)). GaRh and GaPd SCALMS were prepared by adding 50 mL of distilled water to the Ga-decorated support followed by adding an aqueous solution of the ammonium-tetrachloropalladate ($1 \text{ mg}_{\text{Pd}} \text{ mL}^{-1}$) for GaPd and an aqueous solution of rhodium nitrate ($0.94 \text{ mg}_{\text{Rh}} \text{ mL}^{-1}$) in the case of GaRh, respectively. The mixture was stirred for 45 min. The solid was then filtered and washed with distilled water and finally with propan-2-ol. The resulting solid was dried overnight and calcined at 723 K for 2 h (Fig. 1B). The GaPt-SCALMS catalysts were prepared by adding an aqueous solution of hexachloroplatinic acid ($1 \text{ mg}_{\text{Pt}} \text{ mL}^{-1}$) to a suspension of the Ga-decorated support in 50 mL of propan-2-ol. The solvent was evaporated at 323 K in a rotary evaporator and the material was further calcined at 723 K for 2 h.

The bulk metal, supported Pd, Rh and Pt catalysts were synthesized for comparison. For this purpose, nitrate salts of Pd, Rh and Pt were used as the precursor for the metal catalysts. The required amount of the salt was dissolved in 100 mL of distilled water. 10 g of the support material was added to the solution, followed by removal of water in a rotary evaporator at 353 K. The catalysts were reduced *in situ* before use in the catalytic dehydrogenation reaction.

2.2. Characterization of catalysts

The amount of each metal in the prepared catalyst materials was determined by inductively coupled plasma atomic emission spectroscopy (ICP-AES) using a Ciroso CCD (Spectro Analytical Instruments GmbH). The solid samples were digested in 3:2 volumetric ratio of concentrated $\text{HCl}:\text{HNO}_3$ using microwave heating up to 523 K for 40 min. The instrument was calibrated for Rh (443.489 nm), Pd (340.458 nm), Pt (214.423 nm) and Ga (417.206 nm) with standard solutions of the elements prior to the analysis.

CO_2 and NH_3 TPD were performed using a Thermo Scientific TPDRO 1100 instrument equipped with a TCD sensor. Approximately 300 mg of material was used for each of the measurements. For CO_2 TPD, the material was dried at 773 K for 30 min under He flow followed by CO_2 exposure at 313 K for 60 min. The sample was flushed for 90 min after CO_2 exposure. Similarly, for NH_3 TPD the sample was dried at 773 K under He flow followed by NH_3 exposure at 393 K followed by purging with He for 90 min at 313 K. TPD data were obtained in a temperature range of 303 K to 1023 K with a heating rate of 5 K min^{-1} .

High resolution argon physisorption analyses were performed using an Autosorb-iQ XR sorption instrument (Anton Paar Quantatec). Argon with a purity of 6 N (Air Liquide, AlphagazTM 2) was used as adsorptive, while liquid Argon (Riessner Gase) at its boiling temperature of 87.3 K was used as liquid cryogen. All samples were outgassed for 12 h under turbomolecular pump vacuum at 423 K. The BET equation was applied in between relative pressures of 0.05 and 0.3, as recommended by

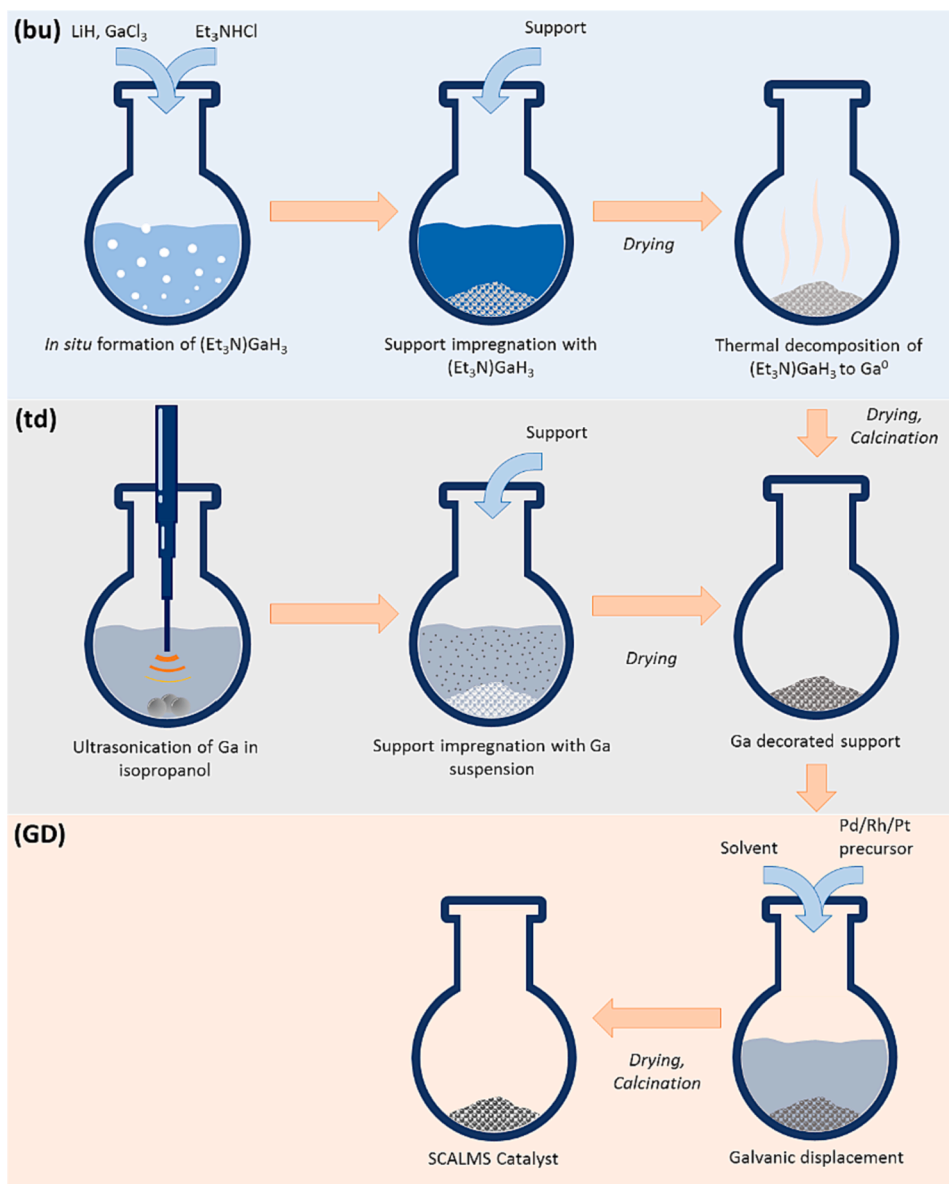


Fig. 1. SCALMS synthesis methodologies: (bu) preparation of Ga decorated support by the bottom-up approach via thermal decomposition of triethylamino-gallane complex; (td) preparation of Ga decorated support by the top-down approach via ultrasonication of Ga nuggets in propan-2-ol; (GD): addition of the active metal by galvanic displacement to produce the final SCALMS catalyst.

IUPAC, using the ASIQwin Software from Anton Paar, Quantatech [27,28].

A Phenom™ XL desktop instrument was used to acquire the scanning electron microscopy (SEM) images. All images were acquired using a back scattered detector (BSD) at 15 kV for elemental contrast.

2.3. Catalytic studies in a fixed-bed tubular reactor

The SCALMS materials prepared by the two described synthesis routes and the bulk metal supported catalyst materials were tested for activity in the isobutane dehydrogenation reaction in a continuous-flow laboratory setup (Fig. S1). The setup was equipped with a fixed-bed reactor (quartz glass of length 650 mm and inner diameter of 10 mm) placed inside a tubular split furnace (Carbolite Gero GmbH). 1 g of the catalyst to be tested was weighed out and loaded into the quartz tube reactor. The reactor was heated to the set point ($T_{\text{reactor}} = 723 \text{ K}$) at 10 K min^{-1} . After a stabilization time of 15 min, the catalyst was first pre-treated in a stream of a 20% H_2 in He for 3 h (total flow of

$\sim 80 \text{ mL}_N \text{ min}^{-1}$) to reduce supported Ga-oxide which stemmed from the synthesis. The formation of traces of water, detected by sensitive TGA-MS, confirmed the reduction [15]. For now, we can only assume that reduction to elemental Ga happened at least in the direct surrounding of the added precious metal that activates the added hydrogen during this reduction process. After pretreatment, the reactor was purged with inert gas for a period of 30 min. Isobutane dehydrogenation was conducted under a flow of $10.7 \text{ mL}_N \text{ min}^{-1}$ of isobutane (99.95% purity, Air Liquide) diluted in $71 \text{ mL}_N \text{ min}^{-1}$ of inerts (90% He + 10% Ar; 99.996% purity, Linde Gas).

2.4. Online analysis of the reaction products

The product stream was analyzed using a Bruker 456 gas chromatography (GC) instrument connected online and equipped with a GC-Gaspro column ($30 \text{ m} \times 0.320 \text{ mm}$) and a flame ionization detector (FID). Mole fractions of compound i (x_i) were calculated from peak areas and calibration factors, which were determined independently for every

substance. The conversion of isobutane ($X_{isobutane}$), the productivity and the selectivity towards butenes ($S_{butenes}$) were calculated as follows:

$$X_{isobutane} = \frac{X_{isobutane,0} - X_{isobutane}}{X_{isobutane,0}} \bullet 100\% \quad (1)$$

$$S_i = \frac{x_i}{X_{isobutane,0} - X_{isobutane}} \bullet 100\% \quad (2)$$

$$S_{butenes} = S_{isobutylene} + S_{1-butene} + S_{cis-2-butene} + S_{trans-2-butene} \quad (3)$$

$$Productivity = \frac{F_{n,isobutane,0} \cdot \left(\frac{X_{isobutane}}{100}\right) \cdot \left(\frac{S_{propene}}{100}\right) \cdot MW_{butene}}{\left(\frac{m_M}{MW_M}\right)} \quad (4)$$

with x_i being the mole fraction and $x_{i,0}$ being the initial mole fraction of compound i as calculated based on online-GC analysis, $F_{n,isobutane,0}$ being the total mole flowrate of propane in the feed and MW_{butene} being the molar weight of butane and m_M being the mass of the noble metal in the bed.

3. Results and discussion

3.1. Catalyst characterization

First, the metal contents of all catalyst materials obtained either in the described bottom-up (bu) or in the described top-down (td) procedure have been determined by ICP-AES. These data are summarized in Table 1 together with the data for the prepared monometallic reference catalyst materials.

The gallium content achieved by the bottom-up approach (thermal decomposition of triethylamino-gallane) was slightly higher than that achieved by the top-down approach (ultrasonication of gallium nugget). The concentration of the active metal with respect to the gallium amount was chosen such that the active metal was fully soluble in gallium under reductive reaction conditions, with the goal to establish a single liquid phase catalytic system.

The top-down route produced gallium droplets with a wide size range and most of these remain on the outer surface of the support. The distribution of gallium on the support is visible in the scanning electron microscopy images in Fig. 2. The alumina support material appears as ensemble of small crystallites with a lot of edges and corners. In the td-catalysts the gallium droplets are clearly visible on the surface of the catalyst and seem to aggregate more on the edges and corners of the support surface. This could explain why the gallium loading is lower in td-SCALMS as we are limited by the amount of gallium that merely the outer surface of the support can accommodate. In the bottom-up approach, in contrast, the triethylamino-gallane precursor enters the porous network of the support and decomposes within the pore structure. Therefore, the scanning electron microscopy images of bu-SCALMS materials show only few Ga droplets at the outer surface of the support (Fig. 3).

The surface chemistry of all catalysts and the bare support materials

Table 1
Composition of SCALMS catalyst tested in this study as identified by ICP-AES.

Catalyst	Ga wt.-%	M _{active} wt.-%	Ga:M _{active} mol _{Ga} mol _M ⁻¹
GaPd_bu	3.83	0.08	73
GaPd_td	3.32	0.08	66
Pd	–	0.14	–
GaRh_bu	3.89	0.04	143
GaRh_td	3.28	0.04	124
Rh	–	0.17	–
GaPt_bu	3.93	0.14	78
GaPt_td	2.43	0.07	97
Pt	–	0.32	–

under investigation was analyzed by CO₂- and NH₃-TPD to see if the synthesis process had any effect on the concentration of the acidic and basic sites of the support (summarized in Table 2). CO₂-TPD of the bare support indicated a basic site concentration of 94 μmol g⁻¹. NH₃-TPD of the bare support indicated an acidic site concentration of 480 μmol g⁻¹. The catalysts synthesized by the top-down approach showed little difference in the basic site concentration compared to the bare support material. Since ultrasonication is carried out in propan-2-ol, this small change in basic site concentration is likely due to the metal addition. In case of the catalysts synthesized by the bottom-up method, however, we observed a significant drop in the basic site concentration for all prepared catalysts. All catalysts had an acidic site concentration in the range of 480 – 580 μmol g⁻¹. No clear dependencies on the catalyst preparation methods can be seen here.

Overall, the top-down approach for Ga deposition on the support does not modify the support surface chemistry.

3.2. Physisorption analysis

To compare the effect of the two synthesis routes on the texture of the system, gas sorption measurements using argon at 87 K were performed for the unmodified Al₂O₃ support, the support after introduction of gallium and the SCALMS catalyst after introduction of the active metal. The support material was analyzed after Ga deposition and after platinum deposition for both, the bottom-up and the top-down preparation method. Using physisorption analysis, only the Pt based SCALMS were investigated. The isotherms are depicted in Fig. 4(a and b). All isotherms are referenced solely by the weight of the Al₂O₃, excluding the weight of Ga and Pt, which was obtained from ICP-AES.

The isotherms represent a type IVa isotherm indicative of mesoporosity according to the IUPAC classification [27]. This type of isotherm is indicative of non-restricted, openly accessible mesopores with size >4 nm [27]. The adsorption hysteresis reveals two regions (between 0.40 < $\frac{p}{p_0}$ < 0.75, the other in between 0.75 < $\frac{p}{p_0}$ < 0.99) resembling different pore systems, whereby the adsorption isotherm does not exhibit a plateau indicating the presence of larger meso/macropore which could not be completely filled by pore condensation in the adsorption experiment.

The impregnation of the support with gallium using the top-down approach (Ga_td) and the successive galvanic displacement of Pt (GaPt_td) does not lead to a visible change in the isotherm. As expected, the deposition of gallium using the bottom-up approach (GaPt_bu) significantly decreased the adsorbed volume. Galvanic displacement of Pt on this material (GaPt_td) in turn leads to an increase in adsorbed volume, not fully restoring the initial uptake by the unmodified support.

The pore size distribution (PSD) and volume distributions are calculated from the adsorption branch by applying a dedicated nonlocal density functional theory (NLDFT) method, assuming an oxidic surface and cylindrical pore geometry. Furthermore, the applied adsorption branch NLDFT kernel takes quantitatively the delay in condensation due to metastable pore fluid correctly into account. The pore size analysis for the unmodified support (Al₂O₃) and the chemically prepared support impregnated solely with gallium (Ga_bu), as well as the one with gallium and platinum (GaPt_bu), is shown in Fig. 4(d and f). The analysis for the top-down synthesis is shown in Fig. 4(c and e). The obtained textural parameters for all materials are listed in Table 3.

The unmodified support predominantly shows a primary mesoporous system with a narrow PSD ranging from pore diameters of 4.3 nm to 8.0 nm centered at ca. 5.7 nm exhibiting a cumulative pore volume of 0.16 cm³ g⁻¹. Additionally, a secondary system, consisting of pores larger than 8.0 nm, is present with a cumulative pore volume of 0.07 cm³ g⁻¹ in between 8 and 45 nm. These larger mesopores fill with capillary liquid at higher relative pressures, the largest ones will fill close to saturation pressure. As already indicated in the previous section, the lack of a plateau in the measured isotherms is based on the presence

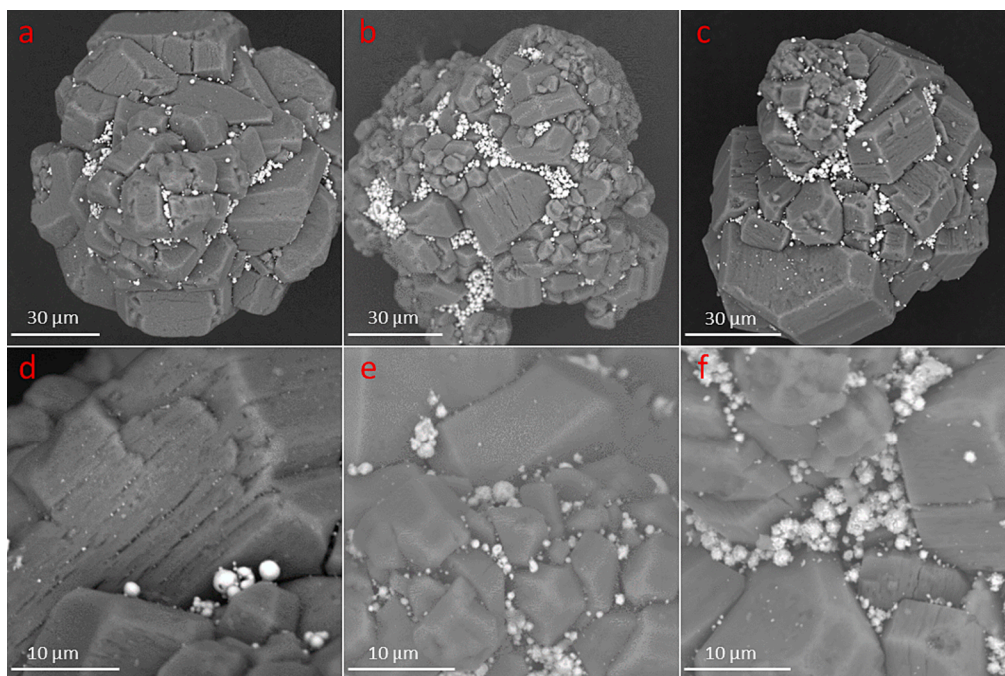


Fig. 2. SEM images of the td-SCALMS catalysts prepared by ultrasonication at lower magnification (a-c) and at higher magnification (d-f) of GaPd (a,d), GaRh (b,e) and GaPt (c,f).

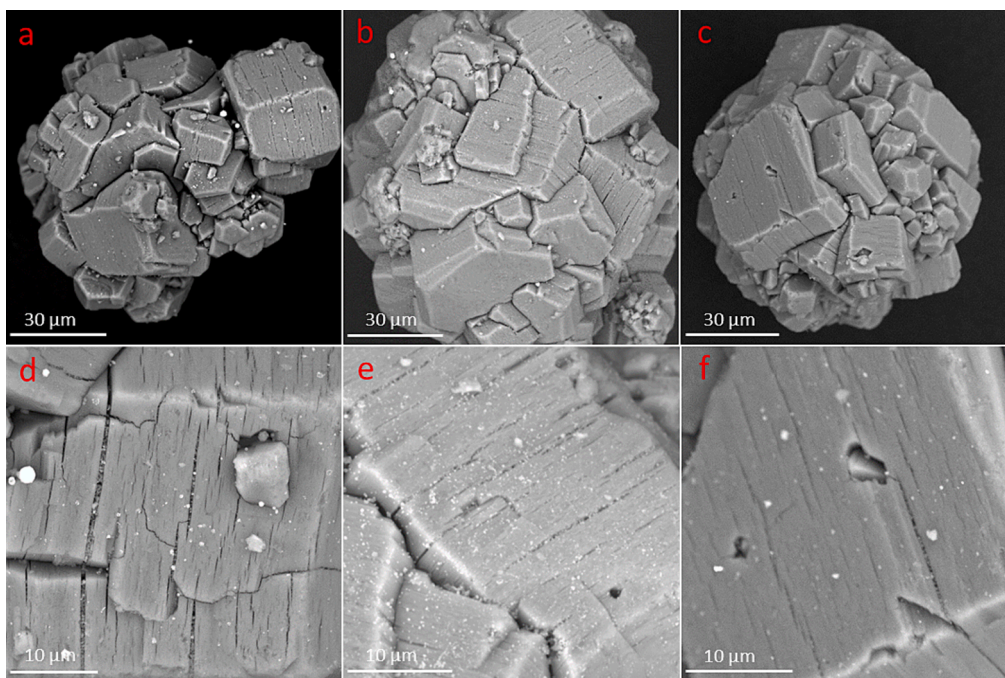


Fig. 3. SEM images of the bu-SCALMS catalyst prepared by chemical route at lower magnification (a-c) and at higher magnification (d-f) of GaPd (a,d), GaRh (b,e) and GaPt (c,f).

of these large mesopores, which extend the hysteresis towards the highest measured relative pressure. The primary mesoporous system centered around ca. 5.7 nm is corresponding to the first segment/step in the adsorption/desorption isotherm, the second system to the second segment. Due to the lack of a reversible plateau, it can be concluded that most likely larger meso- and macropores are additionally present in the unmodified support, which are not resolved through physisorption. Therefore, a mercury porosimetry analysis has been performed, showing that the second meso-/macroporous system extends to about 100 nm (Fig. S7). Additional intraparticle porosity was not observed. According

to the mercury intrusion data, the mesopore volume in between 8 and 45 nm constitutes with >90% the majority of the second porous system. We conclude thereby that the porosity resolved by physisorption is sufficient to draw reliable conclusions representative for the whole interparticle porosity.

The deposition of gallium via the bottom-up method leads to a reduction in pore volume, which is 22% for the pores <8 nm and 23% for the pores 8 nm < d_p < 45 nm and thus similarly distributed over the PSD present. This decrease cannot be fully explained by the occupation of pore volume by deposited gallium. The total volume of elemental

Table 2Acidic and basic site concentration obtained from CO₂/NH₃-TPD analysis.

Catalyst	Concentration of basic sites $\mu\text{mol g}^{-1}$ [†]	Concentration of acidic sites $\mu\text{mol g}^{-1}$ [†]
Al ₂ O ₃	94	480
GaPd_bu	48	580
GaRh_bu	63	456
GaPt_bu	17	588
GaPd_td	120	472
GaRh_td	117	534
GaPt_td	92	517

[†] Error margin of $\pm 10\%$.

gallium, calculated using the weight percentage on the support, accounts with $0.01 \text{ cm}^3 \text{ g}^{-1}$ just for 19% of the reduction in total pore volume. Introduction of gallium also reduces the pore diameter, with its mode being decreased from 5.7 nm to 5.5 nm, its mean (omitting pores larger than 8 nm) being decreased from 5.9 nm to 5.6 nm. The decrease of the pore diameter shows that the triethylamino-gallane complex at least partly penetrates the small mesopores. Interestingly, after addition of Pt via galvanic displacement the pore volume increases from $0.13 \text{ cm}^3 \text{ g}^{-1}$ to $0.15 \text{ cm}^3 \text{ g}^{-1}$ for the mesopores $< 8 \text{ nm}$ and from $0.07 \text{ cm}^3 \text{ g}^{-1}$ to $0.08 \text{ cm}^3 \text{ g}^{-1}$ for the larger mesopores up to 45 nm. For both systems, the pore volume is reduced by 10% compared to the pristine support, comparing to 22% and 23% reduction for the support just loaded with gallium. The introduction of Pt thus increases the available pore volume.

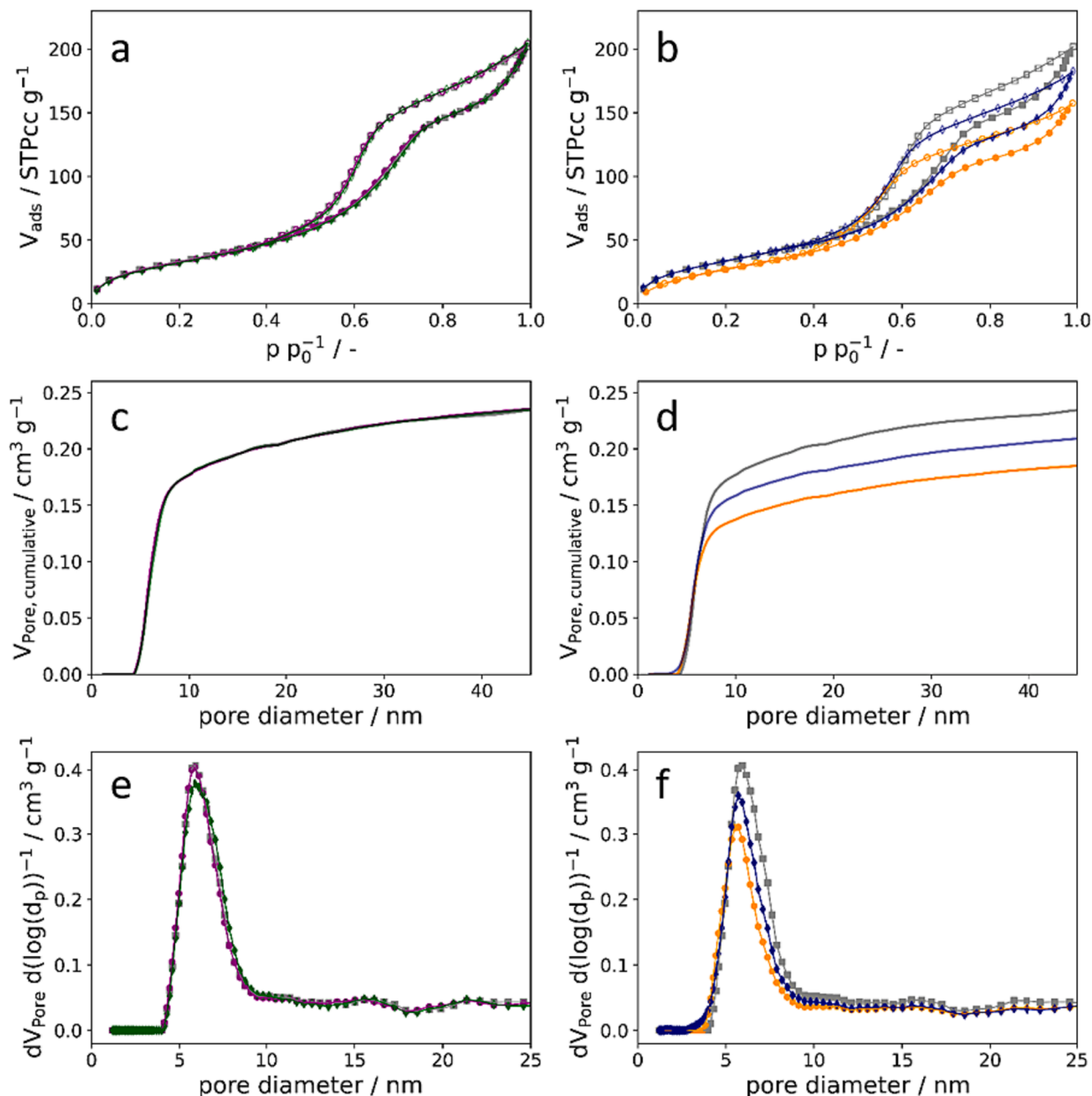


Fig. 4. Argon isotherms at 87.3 K (a,b), NLDFT cumulative (c,d) and differential (e,f) pore size distributions (obtained from the adsorption branch) of catalyst material synthesized by the top-down approach (a,c,e) and bottom-up approach (b,d,f). The materials tested were unmodified aluminum oxide support (Al₂O₃, gray), the support impregnated with gallium via the top-down approach (Ga_td, purple) and bottom-up approach (Ga_bu, orange) and the corresponding catalyst after galvanic displacement (GaPt_td, Green, and GaPt_bu). The adsorption branch of the isotherm is shown with full symbols, the desorption branch with empty symbols (Al₂O₃: square; Ga on support: sphere; GaPt on support: diamond). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3

Textural parameters obtained by NLDFT analysis using argon isotherms at 87 K measured on the unmodified aluminum oxide support (Al_2O_3), as well as the gallium impregnated support for both synthesis conditions ($\text{Ga}_{\text{bu}}/\text{Al}_2\text{O}_3$, $\text{Ga}_{\text{td}}/\text{Al}_2\text{O}_3$) and the corresponding catalyst after galvanic displacement ($\text{GaPt}_{\text{bu}}/\text{Al}_2\text{O}_3$, $\text{GaPt}_{\text{td}}/\text{Al}_2\text{O}_3$).

Material	$S_{\text{BET}}/\text{m}^2\text{g}^{-1}$	$V_{\text{Pore}, d_p < 45\text{nm}}/\text{cm}^3\text{g}^{-1}$	$d_{p,\text{Mode}}/\text{nm}$
Al_2O_3	115	0.25	5.70
Ga_{td}	116	0.26	5.70
GaPt_{td}	114	0.26	5.70
Ga_{bu}	98	0.19	5.51
GaPt_{bu}	116	0.23	5.51

The mode and mean pore sizes are not changed by galvanic displacement. The total reduction in pore volume is still larger than what can be attributed to gallium. This is indicative of pore blocking resulting in isolated regions that are no longer accessible. Fig. 5 shows a schematic illustration of the possible mechanism by which pore blocking would result in isolated regions even in an interconnected pore network.

Suspending the support in the mixture of propan-2-ol and the aqueous solution of hexachloroplatinic acid could lead to agglomeration of gallium droplets. Upon drying, these droplets may be deposited at different positions on the support leading on average to less blocking. Moreover, hexachloroplatinic acid can easily etch the native oxide film on gallium droplets further increasing the mobility and coalescence of the droplets.

We conclude that the td-method that uses ultrasonication to deposit gallium droplets does not penetrate the mesoporous system, since the droplet size obtained via this method is too large. The chemical synthesis achieves the pore penetration even into small mesopores with diameters centered around 5.7 nm. Here, Ga droplets could lead to pore blocking and the observed decrease in the accessible pore volume.

3.3. Catalytic results in isobutane dehydrogenation (i-BDH)

Isobutane dehydrogenation was carried out in a quartz, tubular reactor. The catalysts are compared on basis of productivity Eq. (4) which is calculated on active metal basis and considering all isomers of butene i.e., but-1-ene, isobutylene, *cis*-but-2-ene and *trans*-but-2-ene.

Among the bulk metal catalyst materials, $\text{Pd}/\text{Al}_2\text{O}_3$, although active, shows a very low productivity of $49 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$ which drops to $15 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$ after 14 h on stream. $\text{Rh}/\text{Al}_2\text{O}_3$ is the least active catalyst with an initial productivity of $15 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$ and conversion values approaching the blind activity of the reactor. Pt is already used in the industry as a dehydrogenation catalyst and unsurprisingly $\text{Pt}/\text{Al}_2\text{O}_3$ showed significantly higher initial productivity ($185 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$) compared to $\text{Pd}/\text{Al}_2\text{O}_3$ and $\text{Rh}/\text{Al}_2\text{O}_3$.

Interestingly, in all cases, the bimetallic SCALMS catalysts, irrespective of their synthesis route, showed significantly higher productivities than their bulk metal reference counterparts (Fig. 6). Given the much lower catalyst interface compared to the bulk metal catalysts due to the relatively large droplet size within the SCALMS materials, this general finding is highly remarkable. SCALMS catalyst prepared by the bottom-up approach had 1.5 to 3 times higher activity than the respective catalyst prepared by ultrasonication. We assign this difference in activity to the smaller particle sizes obtained by the bottom-up approach (compare Figs. 2 and 3). From our SEM studies it seems that the bottom-up particles are smaller by a factor of approx. 2. This would result in a higher specific surface by a factor of 4, which is in line with the activity difference observed. However, all catalysts deactivated to varying degrees over the course of the experiment. $\text{GaPd}_{\text{bu}}/\text{Al}_2\text{O}_3$ has an initial productivity of $407 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$ which reduces to about $232 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$ after 14 h on stream. $\text{GaPd}_{\text{td}}/\text{Al}_2\text{O}_3$ has lower initial and final (14 h time-on-stream) productivities of $176 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$ and $70 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$, respectively. Interestingly, the productivity of GaRh SCALMS was much higher approaching that of GaPt SCALMS catalysts. The initial productivities of $\text{GaRh}_{\text{td}}/\text{Al}_2\text{O}_3$ and $\text{GaRh}_{\text{bu}}/\text{Al}_2\text{O}_3$ were $260 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$ and $832 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$, respectively. Note, that Rh is not a very active catalyst for alkane dehydrogenation as previously found for propane dehydrogenation [14,29] and here confirmed for isobutane dehydrogenation. The relative productivity gain between the bulk metal, Rh nanoparticle catalyst and its SCALMS counterpart is highest among the here studied materials for the Rh system. Both GaPt SCALMS catalysts - prepared by ultrasonication (top-down) as well as by thermal decomposition of triethylamino-gallane (bottom-up) approach - outperformed the bulk metal $\text{Pt}/\text{Al}_2\text{O}_3$. While $\text{GaPt}_{\text{td}}/\text{Al}_2\text{O}_3$ had an initial productivity of $268 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$, $\text{GaPt}_{\text{bu}}/\text{Al}_2\text{O}_3$ showed a maximum productivity of $963 \text{ g}_{\text{butenes}} \text{ mol}_{\text{M}}^{-1} \text{ min}^{-1}$.

The product distribution for the different catalysts close to 5 h time on stream is shown in Fig. 7 along with the conversion that was observed at this point. $\text{Rh}/\text{Al}_2\text{O}_3$ and $\text{Pd}/\text{Al}_2\text{O}_3$ have been excluded from this comparison as the conversion values of the respective experiments were extremely low (below 0.4% and 0.8% for Rh and Pd respectively), just above the blind activity of the reactor (for details see Supporting Information Figs. S2 – S4). The product distribution shows the C1-4 hydrocarbon fractions. All catalysts had a high selectivity to isobutylene exceeding 80%. The amount of cracked products was <10%. While the SCALMS catalyst prepared by bottom-up method consistently showed a higher selectivity, the conversion levels differed for the ones prepared by the bottom-up method and therefore a proper comparison at the same conversion cannot be made.

$\text{Pt}/\text{Al}_2\text{O}_3$ has a high average selectivity towards isobutylene. However, it also produces a lot more cracked products. $\text{Pt}/\text{Al}_2\text{O}_3$ also has the

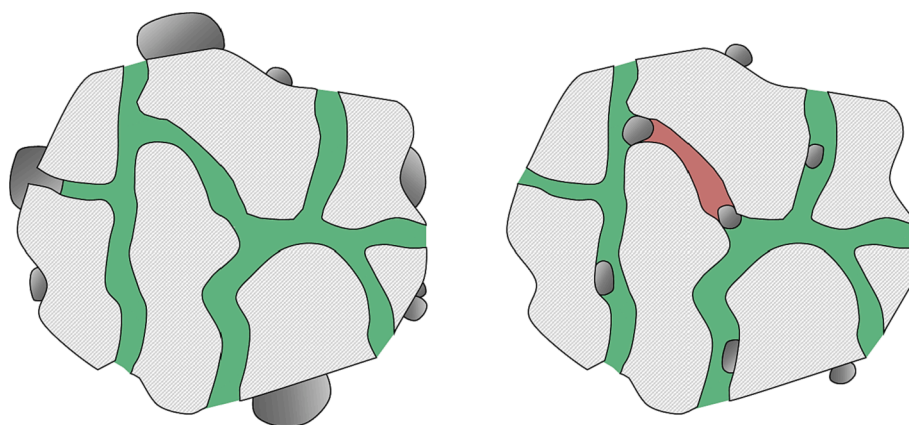


Fig. 5. Schematic illustration of the expected position of the liquid metal droplets for SCALMS prepared by the top-down (a) and bottom-up (b) synthesis approach. The interconnected pore network is represented in green, and the blocked pores are shown in red.

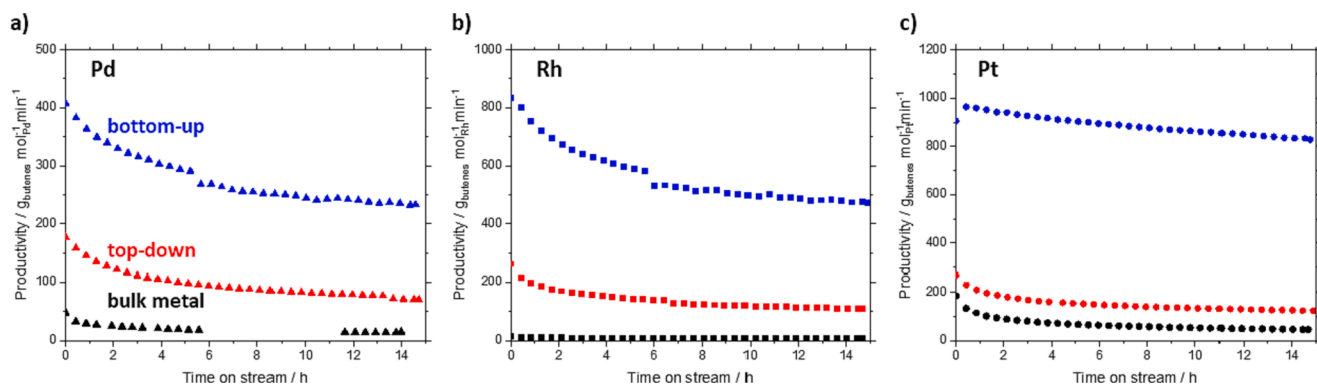


Fig. 6. Comparing productivity obtained by bulk metal catalyst (black) and Ga-SCALMS catalysts prepared by ultrasonication (td, red) and chemical route (bu, blue) with (a) Pd as active metal (triangle) (b) Rh as active metal (square) and (c) Pt as active metal (circle) in isobutane dehydrogenation at 723 K and 1.2 bara. Reaction conditions: 1 g catalyst; $F_{V,He+Ar} = 71 \text{ mL}_N \text{ min}^{-1}$, $F_{V,isobutane,0} = 11 \text{ mL}_N \text{ min}^{-1}$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

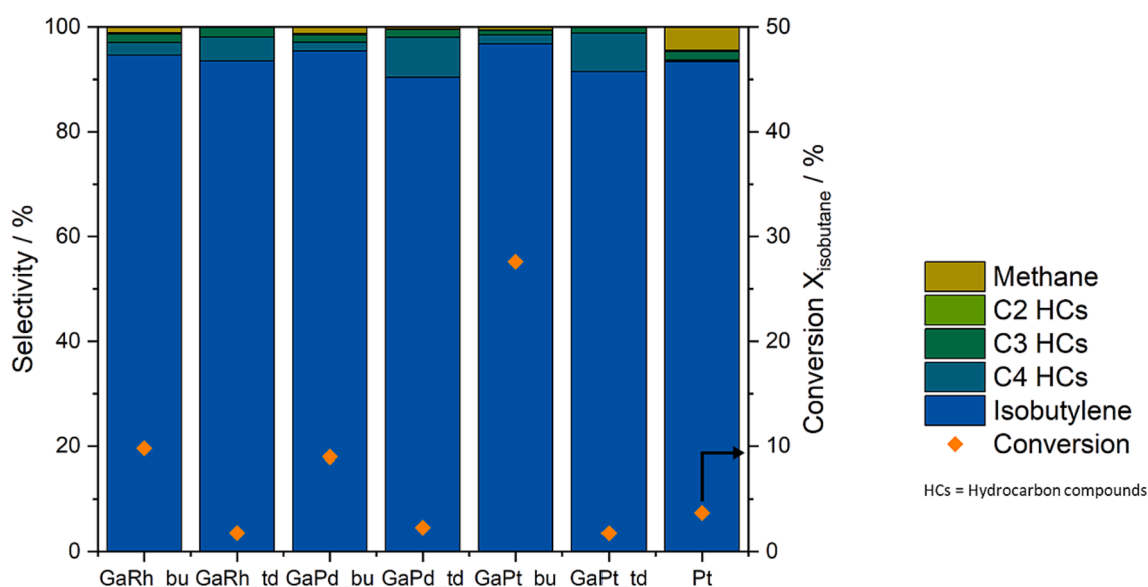


Fig. 7. Product gas distribution as obtained from online-GC at 5 h time on stream and the conversion of isobutane at this point. Reaction conditions: see Fig. 6.

highest deactivation (see Table 4). Deactivation was measured as percentage of activity loss after 14 h on stream compared to maximum initial activity. After 14 h on stream the catalytic activity of Pt/Al₂O₃ decreases by nearly 75%. In fact, this catalyst already deactivates by more than 50% after just 2 h time-on-stream.

Overall, the bulk metal catalysts deactivated the most. Even Pd and Rh, which had very low initial productivity, deactivated by 60–70%. SCALMS catalysts prepared by the top-down approach deactivated by 50–60%. In comparison, catalysts prepared by the bottom-up method showed lower deactivation with GaPt_{bu} retaining nearly 85% of its initial activity after 14 h despite its very high catalytic activity of the system (see Fig. 6). Sintering or agglomeration of Ga droplets can be

Table 4

Percentage of deactivation of the tested catalyst materials after 14 h time on stream.

Active metal, M	Deactivation %		
	Ga-M _{bu}	Ga-M _{td}	M/Al ₂ O ₃
Pd	42.9	59.5	68.2
Rh	43.4	58.1	59.8
Pt	14.1	53.2	74.8

Reaction conditions: see Fig. 6.

excluded as origin of deactivation. We have studied GaPt before and after PDH by means of SEM-EDX and found no difference in the size distribution [13].

Note, that the here described results in isobutene dehydrogenation show a marked difference to earlier published results using GaPt_{td}/Al₂O₃ and GaPt_{bu}/Al₂O₃ for propane dehydrogenation at 823 K [13]. While for propane dehydrogenation similar activities were found for GaPt_{td}/Al₂O₃ and GaPt_{bu}/Al₂O₃, this difference is larger for isobutane dehydrogenation. One possible reason for this difference is that at higher temperature of 823 K, the structural differences caused by top-down and bottom-up synthesis will fade out due to the harsher conditions required for propane dehydrogenation.

Despite the slightly lower activity of the resulting catalyst material, the top-down approach offers a simple and scalable synthesis protocol. It is also clear from our SEM images that the method should be further improved by optimizing the ultrasonic parameters to obtain smaller alloy droplets. This would not only increase the specific surface area of the catalytically active alloy interface, but also make much better use of the porous support. Indeed, it is surprising that such high catalytic activities can be achieved with the top-down approach, where the active alloy interface is comparatively small. This suggests that the interfacial specific activity of the different SCALMS materials is very high and by increasing the dispersion of the alloy droplets in the porous support, the

activity of the whole catalyst material can be further improved. Achieving this is an important focus of our ongoing research activities in this field.

4. Conclusions

In the current study, we present an in-depth comparison of two different synthesis protocols for SCALMS materials, the top-down and the bottom-up preparation method. Both methods have been described in previous work but have not yet been directly compared for the same material systems and in the same catalytic dehydrogenation reaction. For this purpose, GaPd, GaRh and GaPt SCALMS catalysts were prepared by the two methods and tested at 723 K for catalytic activity in isobutane dehydrogenation reaction. For comparison, bulk metal catalysts using Pd, Rh and Pt as active component were also tested.

As expected from our previous work, GaPd, GaRh and GaPt SCALMS were found significantly more active than Pd and Rh nanoparticles on the same alumina support. More interestingly, all SCALMS catalysts showed lower deactivation than their bulk metal counterparts. In all cases SCALMS catalysts synthesized by the bottom-up method resulted in higher activity than SCALMS catalysts synthesized by the top-down method.

The difference in activity is most likely due to differences in the size and distribution of the catalyst alloy droplets on the support, as shown by our SEM studies. SCALMS materials synthesized by the bottom-up method have a much higher degree of pore penetration. The ultrasonic treatment, in contrast, results in large alloy droplets that are mainly located on the outer surface of the support grains. It is noteworthy that despite the poor utilization of the porous support in the top-down synthesis, the resulting catalytic material still outperforms the classical bulk metal catalysts. This speaks for a highly active catalytic interface and opens much room for optimizing the catalytic activity of SCALMS materials by better dispersion of the alloy droplets.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jakob Soellner, Sharanya Nair reports financial support was provided by German Research Foundation. Narayanan Raman, Nicola Taccardi reports financial support was provided by European Research Council.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Experimental setup, isobutane conversion with SCALMS catalyst and pure metal catalyst, post-run SEM images. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2023.146081>.

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